

CARTAGENA, Colombia

WMO: 800220

Lat: 10.442N

Lon: 75.513W

Elev: 4

StdP: 14.69

Time Zone: -5.00 (W05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	73.8	75.0	69.4	108.0	78.4	69.9	110.0	78.7	21.5	83.7	19.9	83.8	3.1	20	0.398

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	10.2	91.7	81.7	91.2	81.5	90.0	80.9	83.3	89.2	82.7	88.4	82.1	87.8	9.1	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
81.4	163.0	87.0	80.9	160.5	86.6	80.6	159.1	86.4	47.3	90.4	46.7	89.3	46.0	88.3	87.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.8	16.4	14.2	DB	69.9	95.4	3.6	2.6	67.3	97.3	65.1	98.8	63.1	100.3	60.5	102.2	(4)
			WB	68.0	85.4	3.6	1.2	65.4	86.2	63.3	86.9	61.3	87.5	58.7	88.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	83.1	81.4	81.6	82.2	83.4	83.9	84.1	84.0	84.0	83.3	83.0	82.6		
		DBStd	1.94	1.36	1.58	1.51	1.57	1.74	1.80	1.65	1.82	1.97	1.92	1.80	1.55	(7)
		HDD50	0	0	0	0	0	0	0	0	0	0	0	0		
		HDD65	0	0	0	0	0	0	0	0	0	0	0	0		
		CDD50	12092	973	885	998	1001	1051	1024	1054	1055	1020	1032	989	1010	(10)
		CDD65	6617	508	465	533	551	586	574	589	590	570	567	539	545	(11)
		CDH74	74480	5022	4609	5473	6208	6956	6935	6982	7040	6833	6533	6055	5834	(12)
(13)		CDH80	26362	1459	1351	1610	2110	2631	2762	2697	2751	2696	2352	2049	1894	(13)
(14)	Wind	WSAvg	6.1	7.1	7.9	8.4	7.2	5.4	5.1	5.8	5.4	5.0	4.9	5.0	6.1	(14)
(15)	Precipitation	PrecAvg	43.4	0.2	0.0	0.1	1.0	3.9	4.1	4.3	5.3	5.9	9.8	5.8	1.7	(15)
(16)		PrecMax	96.0	4.3	0.4	2.0	4.3	11.5	11.1	14.9	16.9	14.5	23.8	22.8	8.2	(16)
(17)		PrecMin	15.8	0.0	0.0	0.0	0.0	0.2	0.4	0.2	0.9	1.7	2.0	1.4	0.0	(17)
(18)		PrecStd	17.4	0.7	0.1	0.4	1.2	2.8	2.6	3.4	3.4	2.9	5.1	4.1	2.3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	89.9	90.0	90.0	91.5	91.8	93.0	92.8	92.1	93.0	91.8	91.5	91.3		
(20)		0.4% MCWB	78.7	78.4	79.3	81.2	82.0	82.5	81.6	81.9	82.5	82.1	81.6	80.4		
(21)		2% DB	88.1	88.2	88.5	89.7	90.0	91.2	91.1	91.0	91.2	89.9	89.7	89.6		
(22)		2% MCWB	78.7	78.4	79.2	80.5	81.2	81.9	81.4	81.6	82.0	81.2	80.9	79.8		
(23)		5% DB	87.4	87.6	87.7	88.2	89.5	89.8	89.8	89.7	89.7	89.2	88.2	88.1		
(24)		5% MCWB	78.6	78.3	79.0	80.1	81.0	81.1	80.8	81.0	81.2	80.9	80.4	79.6		
(25)		10% DB	86.1	86.2	86.3	87.6	88.0	88.9	88.7	88.2	88.2	87.8	87.5	87.5		
(26)		10% MCWB	78.0	77.8	78.4	79.9	80.6	80.8	80.6	80.7	80.7	80.4	80.1	79.4		
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	80.6	80.7	81.6	82.7	83.6	83.7	83.3	83.6	83.7	83.6	82.8	82.4		
(28)		0.4% MCDB	87.2	87.3	86.9	88.4	89.2	89.9	89.3	89.5	90.0	89.3	88.5	87.5		
(29)		2% WB	79.8	79.8	80.4	81.6	82.6	82.7	82.3	82.6	82.8	82.4	81.8	81.2		
(30)		2% MCDB	86.3	86.2	86.3	87.3	88.1	88.9	88.6	88.4	88.6	87.8	87.1	86.7		
(31)		5% WB	79.1	79.1	79.7	80.7	81.7	81.9	81.6	81.8	82.1	81.6	81.0	80.4		
(32)		5% MCDB	85.5	85.5	85.5	86.5	87.1	87.9	87.7	87.4	87.5	86.7	86.1	86.2		
(33)		10% WB	78.4	78.3	79.0	80.0	81.0	81.3	80.9	81.2	81.4	80.8	80.4	79.8		
(34)		10% MCDB	84.5	84.5	84.7	85.5	86.2	86.9	86.8	86.5	86.5	85.7	85.3	85.2		
(35)	Mean Daily Temperature Range	MDBR	11.3	11.2	10.0	9.4	9.4	10.2	10.7	10.3	10.0	9.9	9.7	10.9		
(36)		5% DB MCDBR	12.2	12.2	11.1	10.2	10.5	11.1	11.6	11.1	11.2	11.1	10.6	11.8		
(37)		5% WB MCWBR	5.7	5.5	5.2	4.9	4.9	5.2	5.3	5.2	5.2	5.0	4.8	5.3		
(38)		5% DB MCDBR	11.4	11.1	10.1	9.6	10.0	10.7	11.0	10.6	10.6	10.6	9.9	10.9		
(39)		5% WB MCWBR	5.7	5.6	5.1	4.9	5.0	5.5	5.5	5.5	5.3	5.3	4.9	5.2		
(40)	Clear-Sky Solar Irradiance	taub	0.401	0.434	0.470	0.524	0.485	0.459	0.455	0.455	0.448	0.411	0.402	0.403		
(41)		taud	2.496	2.393	2.292	2.124	2.275	2.356	2.376	2.378	2.369	2.507	2.534	2.516		
(42)		Ebn at Noon	287	281	272	254	259	263	265	269	274	284	285	284		
(43)		Edh at Noon	34	39	44	51	43	39	39	39	40	34	33	33		
(44)	All-Sky Solar Radiation	RadAvg	1778	1895	1868	1795	1629	1692	1753	1727	1628	1486	1459	1600		
(45)		RadStd	44	85	100	101	117	109	108	107	81	82	135	123		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.63	N/A	N/A	+1.28	+0.74	+0.36	N/A	N/A	N/A	N/A	(46)
(47)	Regional (1 neighbor)	+0.58	N/A	N/A	+1.12	+0.58	+0.25	N/A	N/A	+220	+220	(47)

Nomenclature: See separate page